LAB PRACTICAL GUIDE & REFERENCE

14. Backup and Restore (tar & gzip)

Complete step-by-step shell commands, options explanation, and verification procedures for Linux archiving and compression tasks.

Task 1: Create a compressed backup of the Music folder

Task Requirement: *Create a backup of your Music folder (or any folder with at least 3 files) using the tar command and compress it with gzip, saving the output as music_backup.tar.gz in your home directory.*

```
# Change to home directory context and archive Music directory
tar -czvf ~/music_backup.tar.gz -C ~ Music
```

Flag & Parameter Explanation:

- **`-c`:** Create a new archive.
- **`-z`:** Compress the archive using gzip (.gz).
- **`-v`:** Verbose mode; displays each file being processed in real-time.
- **`-f`:** Specifies the archive file name (must immediately precede filename).
- **`-C ~ Music`:** Switches directory to user home (~) before archiving Music, avoiding full absolute paths in the archive.

Task 2: Extract archive into a new directory and verify contents

Task Requirement: *Extract the contents of a tar.gz file (for example, music_backup.tar.gz) into a new directory called restored_music and verify that all files are restored correctly.*

```
# 1. Create the destination directory
mkdir -p ~/restored_music

# 2. Extract into restored_music directory
tar -xzvf ~/music_backup.tar.gz -C ~/restored_music

# 3. Verify extracted contents recursively
ls -lR ~/restored_music
```

Flag & Parameter Explanation:

- **`-x`:** Extract files from an existing archive.

- **`-C ~/restored_music:`** Directs tar to unpack all files into the specified target folder instead of the current working directory.
- **`ls -lR:`** Recursively lists files, permissions, and sizes to ensure complete restoration.

Task 3: Filtered compressed backup using find and --files-from

Task Requirement: *Write a shell command that creates a compressed backup of all .jpg images from your Downloads folder into a file named images_backup.tar.gz, excluding any files larger than 5MB (Hint: Use the find command with size filtering and tar with --files-from).*

```
# Search for .jpg files < 5MB and pipe list directly into tar
find ~/Downloads -type f -name "*.jpg" -size -5M | tar -czvf ~/images_backup.tar.gz -
-files-from=-
```

Flag & Parameter Explanation:

- **`find ~/Downloads:`** Recursively scans the Downloads folder for matching criteria.
- **`-type f:`** Matches regular files only (ignores sub-folders).
- **`-name "*.jpg":`** Matches files ending with .jpg extension (case-insensitive alternative: -iname).
- **`-size -5M:`** Filters for files strictly smaller than 5 Megabytes.
- **`--files-from=-:`** Tells tar to read the file list directly from standard input (pipe |).

Task 4: Simulate restore scenario (selective file recovery)

Task Requirement: *Simulate a restore scenario: delete one file from your original Music folder, then use your music_backup.tar.gz archive to restore only the deleted file without overwriting the rest.*

```
# Step 1: Simulate accidental deletion of a single file
rm ~/Music/song1.mp3

# Step 2: List contents of archive to confirm relative path of file
tar -tzvf ~/music_backup.tar.gz

# Step 3: Restore ONLY the deleted file into the original location
tar -xzvf ~/music_backup.tar.gz -C ~ Music/song1.mp3
```

Flag & Parameter Explanation:

- **`tar -tzvf:`** Lists ('table of contents') archive members without extracting them to inspect paths.
- **`Music/song1.mp3:`** Specifies the exact member path to extract. tar selectively unpacks only this file and leaves all other existing files untouched.